

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section.

Use the multiple choice answer sheet for Questions 1-10.

- 1 Which of the following is equal to $\frac{1}{\sqrt{5}-1}$?
- (A) $\sqrt{5}-1$
- (B) $\frac{\sqrt{5}+1}{4}$
- (C) $\frac{\sqrt{5}-1}{4}$
- (D) $\frac{\sqrt{5}-1}{\sqrt{5}+1}$
- 2 What is the solution(s) to the equation $|2k+1|=k+1$?
- (A) 0 only
- (B) 0 or $\frac{2}{3}$
- (C) 0 or $\frac{-2}{3}$
- (D) 0 or $\frac{-1}{2}$
- 3 A parabola is concave down and its vertex is $(2, 0)$.
Which statement about the discriminant (Δ) of the parabola is correct?
- (A) $\Delta > 0$
- (B) $\Delta = 0$
- (C) $\Delta < 0$
- (D) $\Delta \leq 0$

Multiple Choice (continued)

4 If $f'(x) = 2\cos(5x)$ and c is a real constant, then what is $f(x)$ equal to?

(A) $-\frac{2}{5}\sin(5x) + c$

(B) $\frac{2}{5}\sin(5x) + c$

(C) $-10\sin(5x) + c$

(D) $10\sin(5x) + c$

5 What are the coordinates of the focus of the parabola $(x+3)^2 = -12y$?

(A) $(-3, -3)$

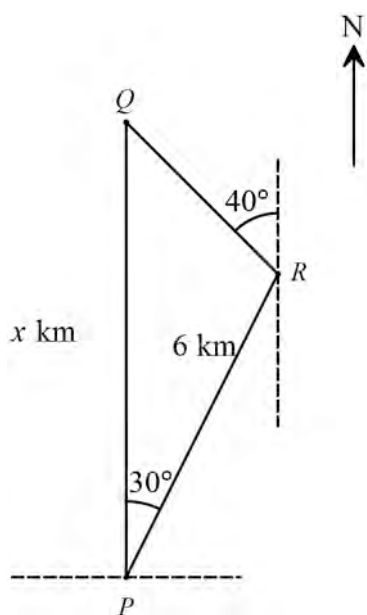
(B) $(-3, 3)$

(C) $(0, -3)$

(D) $(0, 3)$

Multiple Choice (continued)

- 6 A ship leaves a port, P , and sails 6 km on a heading of $N30^\circ E$ to position R . It then heads $N40^\circ W$ until it reaches a port, Q , which is directly north of P .



Which equation represents the distance x km from P to Q ?

- (A) $\frac{x}{\sin 40^\circ} = \frac{6}{\sin 70^\circ}$
- (B) $\frac{x}{\sin 30^\circ} = \frac{6}{\sin 40^\circ}$
- (C) $\frac{x}{\sin 40^\circ} = \frac{6}{\sin 30^\circ}$
- (D) $\frac{x}{\sin 110^\circ} = \frac{6}{\sin 40^\circ}$

Multiple Choice (continued)

7 If $y = 2 \tan(2x)$, then which expression represents $\frac{dy}{dx}$?

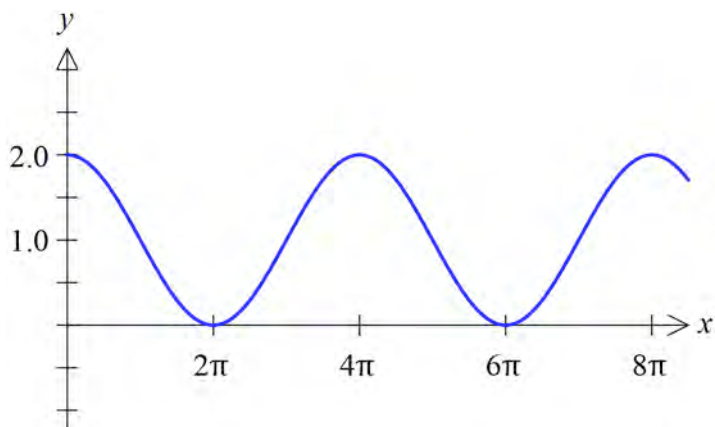
(A) $\frac{1}{\cos^2(2x)}$

(B) $\frac{2}{\cos^2(2x)}$

(C) $\frac{4}{\cos^2(x)}$

(D) $\frac{4}{\cos^2(2x)}$

8 The diagram below shows part of the graph of a circular function.



Which equation represents the graph shown?

(A) $y = 1 + \sin(x)$

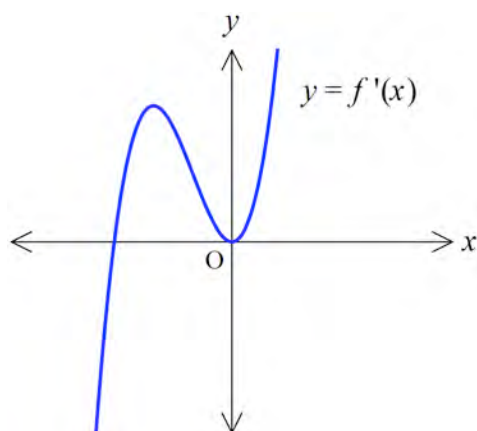
(B) $y = 1 + \sin\left(\frac{x}{2}\right)$

(C) $y = 1 + \cos(x)$

(D) $y = 1 + \cos\left(\frac{x}{2}\right)$

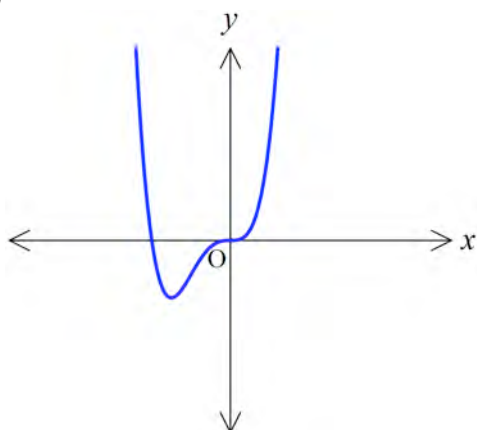
Multiple Choice (continued)

- 9 The graph of a curve $y = f'(x)$, is shown below.

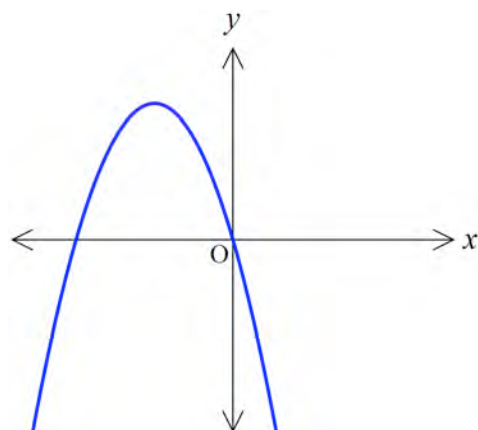


Which one of the following is most likely to be the graph of the function of $f(x)$?

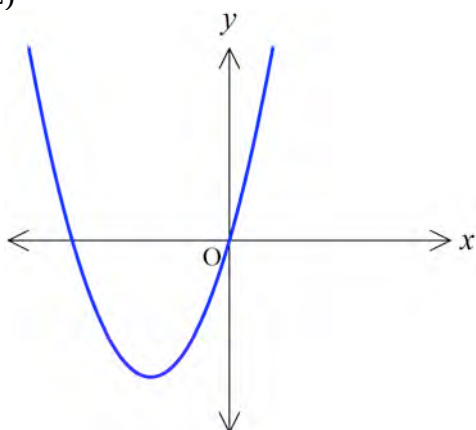
(A)



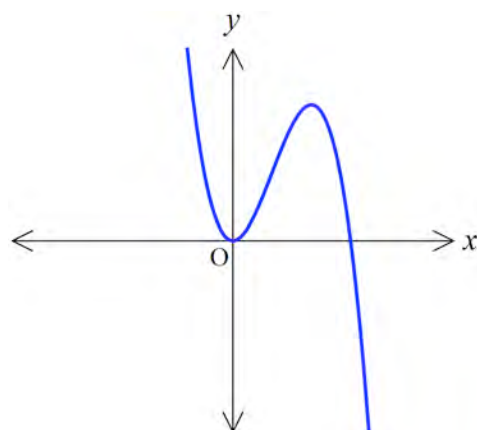
(B)



(C)



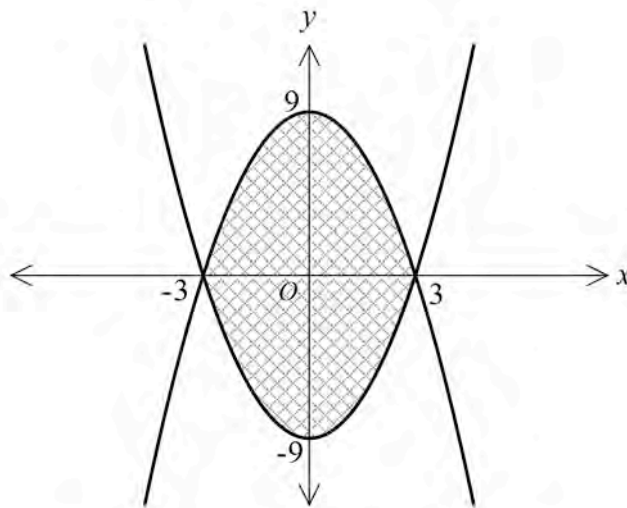
(D)



Multiple Choice (continued)

- 10** The area of the region enclosed between the equations $y = x^2 - 9$ and $y = 9 - x^2$ is shaded in the diagram. Which integral could be used to calculate the shaded area?

- (A) $\int_{-3}^3 2x^2 - 18 dx$
- (B) $2 \int_0^3 18 - 2x^2 dx$
- (C) $\int_{-9}^9 2x^2 - 18 dx$
- (D) $\int_{-9}^9 18 - 2x^2 dx$



Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section.

Answer each question in the appropriate writing booklet. Extra booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a **separate** writing booklet.

Marks

(a) Solve $x^2 - 3 = 3x + 1$. 2

(b) Solve the simultaneous equations 2

$$3x - y = -7$$

$$5x + 2y = 3$$

(c) Differentiate $\frac{2x^3}{4x + 2}$. 2

(d) Evaluate $\int_0^{\pi} \sin 2x \, dx$. 2

(e) Evaluate $\int_0^1 \frac{5}{\sqrt{e^x}} \, dx$. 3

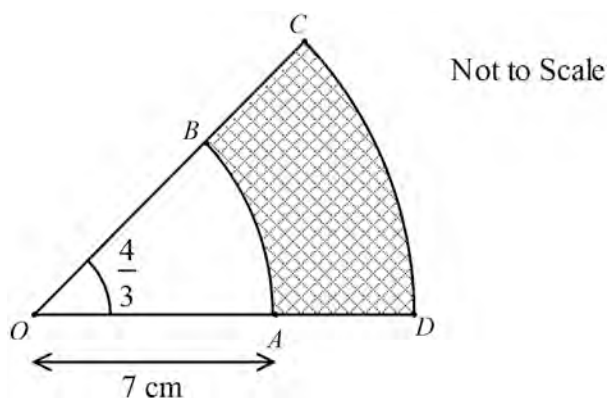
Question 11 continues on page 8

- (f) The angle of a sector in a circle of radius 7 cm is $\frac{4}{3}$ radians, as shown in the diagram.

The points A and B lie on OD and OC respectively and AB is an arc of a circle.

O is the centre of the circle.

The area of the shaded region $ABCD$ is 48 cm^2 .



- (i) Find the distance OD .

2

- (ii) Find the perimeter of the shaded region.

2

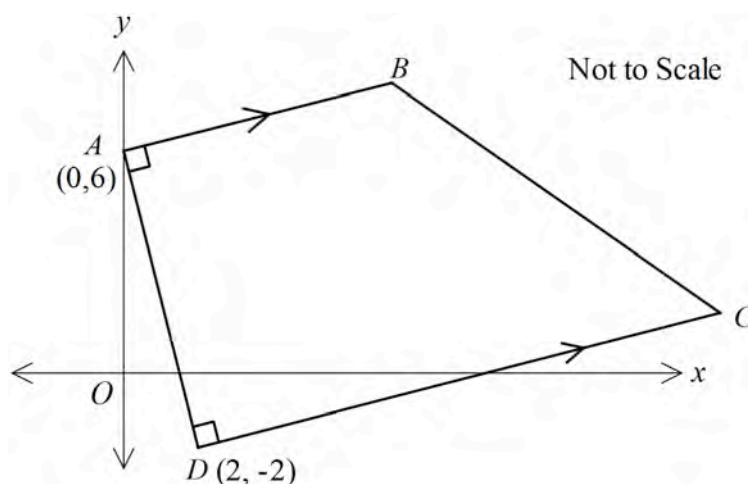
End of Question 11

- (a) Calculate the limiting sum of the infinite geometric series given by

2

$$2 - 1 + \frac{1}{2} \dots$$

- (b)

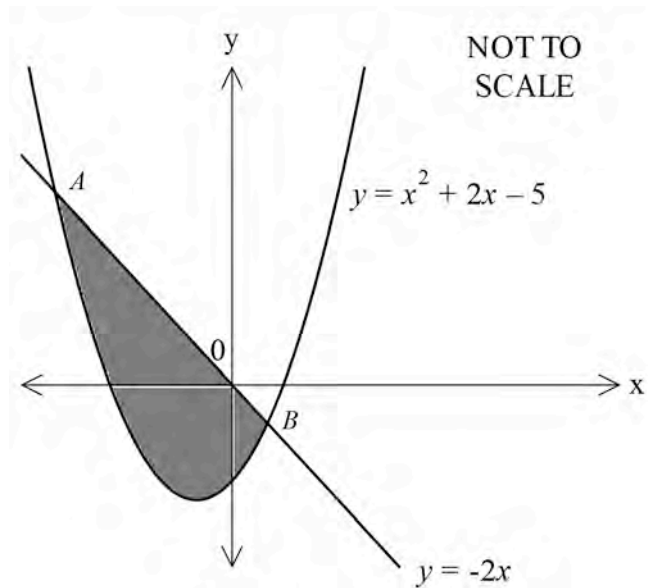


The diagram shows a trapezium $ABCD$ in which AB is parallel to DC and BA is perpendicular to AD . The length of DC is twice the length of AB . The point A is $(0, 6)$ and the point D is $(2, -2)$.

- (i) Show that equation of AB is $x - 4y + 24 = 0$. 2
- (ii) Given that B lies on the line $y = x$, find the coordinates of B . 2
- (iii) Find the area of the trapezium $ABCD$. 2
- (iv) Find the coordinates of C . 2

Question 12 continues on page 10

- (c) The diagram shows the graphs of $y = x^2 + 2x - 5$ and $y = -2x$. These two graphs intersect at point A and point B.



- (i) Find the x -coordinates of the points of intersection A and B. 2
- (ii) Calculate the area of the shaded region. 3

End of Question 12

- (a) For the parabola $8x = 16y - y^2$.
- (i) Find the coordinates of the vertex. 2
 - (ii) Find the coordinates of the focus. 1
 - (iii) Sketch the curve showing all relevant features. 1
- (b) A man buys a new motorcycle. After t months its value $\$V$ is given by $V = 10\,000e^{-pt}$, where p is a constant.
- (i) Find the value of the motorcycle when the man bought it. 1
 - (ii) The value of the motorcycle after 12 months is expected to be $\$4000$.
Calculate the expected value of the motorcycle after 18 months, correct to the nearest dollar. 3
 - (iii) Calculate the age of the motorcycle, to the nearest month, when its expected value will be less than $\$1000$. 2
- (c) A particle moves in a straight line, so that, t seconds after leaving a fixed point O , its velocity, vms^{-1} , is given by $v = \frac{12}{(t+1)^2} - 3$.
- Find:
- (i) an expression for the acceleration of the particle in terms of t . 2
 - (ii) the distance travelled by the particle before it comes to instantaneous rest. 3

End of Question 13

-
- (a) (i) Use one application (two function values) of the trapezoidal rule to find an approximation to
- $$\int_0^2 \sqrt{16-x^2} \, dx.$$
- (ii) Explain whether this approximation is greater than or less than the exact value.
- (b) Consider the function defined by $f(x) = x^3 + 3x^2 - 9x + 5$.
- (i) Find the coordinates of the stationary points of the curve $y = f(x)$ and determine their nature.
- (ii) Find the coordinates of any point of inflexion.
- (iii) Sketch the graph of $f(x) = x^3 + 3x^2 - 9x + 5$ by showing the above information.
- (c) In its first year of production, 6000 mobile phones were sold by a company. Each year after that, sales were 15% more than the previous year's sales.
- (i) Find the sales in the 10th year of production. Express your answer to the nearest ten.
- (ii) Find the total sales in the first 10 years of production. Express your answer to the nearest ten.
- (iii) When will the total profit reach \$1 million if the company made \$10 profit on each sale? (Answer to the nearest whole year).

End of Question 14

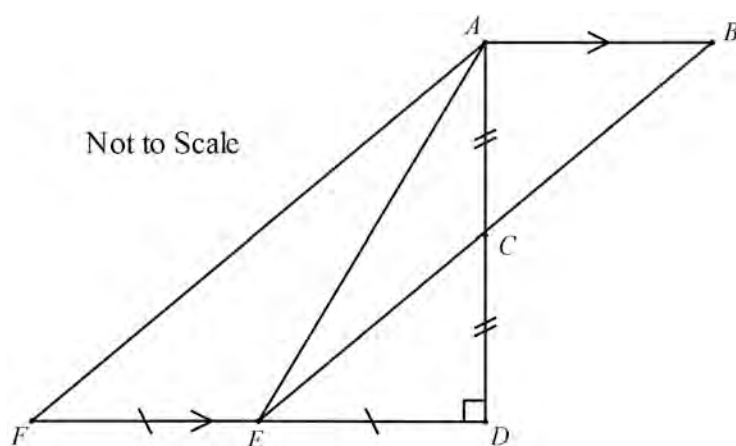
- (a) Solve $\sin x = \cos x$ for $0 \leq x \leq 2\pi$.

2

- (b) Prove the identity $\frac{1}{1 + \tan^2 A} = (1 + \sin A)(1 - \sin A)$.

2

(c)



In the diagram $AB \parallel FD$, ADF is a right-angled triangle, C is the midpoint of AD and E is the midpoint of FD .

- (i) Explain why $\angle CED = \angle ABC$.

1

- (ii) Show that $\triangle CDE \equiv \triangle CAB$.

2

- (iii) Show that $AF = 2BC$.

2

Question 15 continues on page 14

(d) The roots of the quadratic equation $2x^2 + 4x + 5 = 0$ are α and β .

(i) Find the value of $\alpha + \beta$. 1

(ii) Show that $\frac{\alpha^2 + \beta^2}{\alpha\beta} = -\frac{2}{5}$. 2

(iii) The roots of $x^2 + mx + p = 0$ are $\frac{\alpha}{\beta} + 2$ and $\frac{\beta}{\alpha} + 2$. 3

It is also given that $\frac{\alpha^2 + \beta^2}{\alpha\beta} = -\frac{2}{5}$.

Find the values of m and p where m and p are constants.

End of Question 15

- (a) The depth D , in metres, of a liquid stored in a container at time t seconds is given by

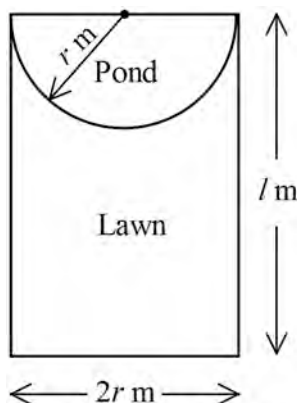
$$D = \frac{t^2 + 1}{e^{2t}}, \quad t \geq 0$$

- (i) Find an expression for the rate at which the depth of the liquid changes. 2
- (ii) Hence, explain whether the depth of the liquid was increasing or decreasing at $t = 10$. 2

- (b) A curve has the equation $y = x^2 \log_e x$, where $x > 0$.

- (i) Find an expression for $\frac{dy}{dx}$. 2
- (ii) Hence, find $\int x \log_e x \, dx$. 2

- (c) A garden is being designed to include a semi-circular pond in a rectangular shaped lawn. The radius of the pond is r metres and the length of the lawn is l metres, as shown in the diagram below.



- (i) Given that the area of the lawn is 400 m^2 , express l in terms of r . 2
 Show that $l = \frac{200}{r} + \frac{\pi}{4}r$.
- (ii) Given that the perimeter of the lawn is P m, show 2
 that $P = \left(\frac{3\pi}{2} + 2 \right)r + \frac{400}{r}$.
- (iii) Given that r and l can vary, find the value of r for which P is a minimum length. 3

HSC Trial Examination 2015
Mathematics - SOLUTIONS

①

SECTION I (10 marks)

- 1 B 6 D
2 C 7 D
3 B 8 D
4 B 9 A
5 A 10 B

SECTION II (90 marks)

Question 11: (15 marks)

a) $x^2 - 3 = 3x + 1$

1 mark factorisation

$\frac{1}{2} \quad x^2 - 3x - 4 = 0$

$(x-4)(x+1) = 0$

$x = 4 \text{ or } -1$

1 mark

b) $3x - y = -7$ ① $\rightarrow y = 3x + 7$

$\frac{1}{2} \quad 5x + 2y = 3$ ②

Sub. $y = 3x + 7$ into ②: $5x + 2(3x + 7) = 3$

$11x + 14 = 3$

$11x = -11$

$x = -1$

1 mark

Sub. $x = -1$ into $y = 3x + 7$
 $= 3(-1) + 7$

$y = 4$

1 mark

Sol'n $\begin{cases} x = -1 \\ y = 4 \end{cases}$

c) $y = \frac{2x^3}{4x+2}$
 $\frac{1}{2}$

$= \frac{u}{v}$

where $u = 2x^3$
 $u' = 6x^2$

; $v = 4x + 2$
 $v' = 4$

Q11

(2)

$$y' = \frac{vu' - uv'}{v^2}$$

$$= \frac{(4x+2) \cdot 6x^2 - 2x^3 \cdot 4}{(4x+2)^2}$$

1 mark

$$= \frac{24x^3 + 12x^2 - 8x^3}{(4x+2)^2}$$

1 mark

$$= \frac{4x^2(6x+3-2x)}{(4x+2)^2}$$

$$y' = \frac{4x^2(4x+3)}{(4x+2)^2}$$

$$\text{or } y' = \frac{4x^3 + 3x^2}{(2x+1)^2}$$

$$\frac{1}{2} \text{ d) } \int_0^\pi \sin 2x \, dx = \left[-\frac{1}{2} \cos 2x \right]_0^\pi$$

1 mark

$$= -\frac{1}{2} (\cos 2\pi - \cos 0)$$

$$= -\frac{1}{2} (1 - 1)$$

$$= 0$$

1 mark

$$\frac{1}{3} \text{ e) } \int_0^1 \frac{5}{\sqrt{e^x}} \, dx = \int_0^1 5e^{-\frac{x}{2}} \, dx$$

$$= 5 \times \left(\frac{+1}{-\frac{1}{2}} \right) \left[e^{-\frac{x}{2}} \right]_0^1$$

$$= -10 \left(\frac{1}{\sqrt{e}} - \frac{1}{\sqrt{e^0}} \right)$$

$$= -10 \left(\frac{1}{\sqrt{e}} - 1 \right)$$

$$= -10 \left(\frac{1}{\sqrt{e}} - 1 \right)$$

3 marks all correct

2 marks correct integration plus correct substitution.

1 mark correct integration.

Q11

③

$$f) A = \frac{1}{2} r^2 \theta$$

$$\text{Area ABCD} = \frac{1}{2} (x+7)^2 \cdot \frac{4}{3} - \frac{1}{2} \cdot 7^2 \cdot \frac{4}{3}$$

$$48 = \frac{2}{3} [x^2 + 14x + 49 - 49]$$

$$72 = x^2 + 14x$$

$$0 = x^2 + 14x - 72$$

$$= (x+18)(x-4)$$

$$x = -18 \text{ or } 4$$

↑

Disregard

 $x > 0$

$$\therefore \text{Sol'n } x = 4 \text{ cm}$$

$$\begin{array}{l} i \\ \frac{1}{2} \end{array} \quad \begin{array}{l} OD = 7 + 4 \\ = 11 \text{ cm} \end{array}$$

2 marks all correct

1 mark - 1 error

$$\begin{array}{l} ii \\ \frac{1}{2} \end{array} \quad P = 4 \times 2 + 7 \times \frac{4}{3} + 11 \times \frac{4}{3} \quad (l = r\theta)$$

$$P = 32 \text{ cm}$$

2 marks all correct

1 mark - showed some understanding of
perimeter being sum of 2 arcs
and 2 radii

Question 12 (15 marks)

$$a) \quad 2 - 1 + \frac{1}{2} \dots \quad r = -\frac{1}{2}, \quad a = 2$$

$$\frac{1}{2} \quad (-1 < r < 1)$$

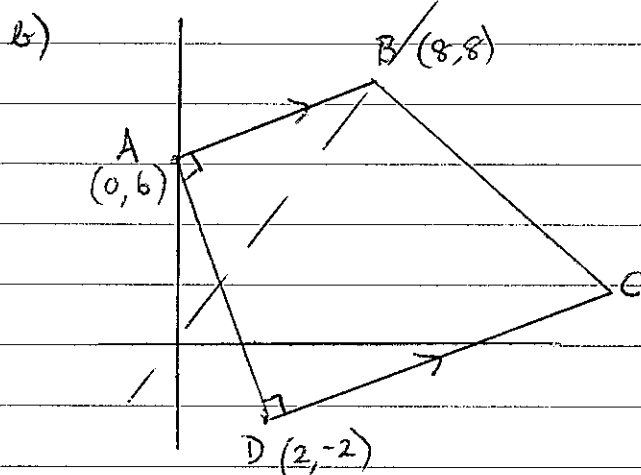
$$S_{\infty} = \frac{a}{1-r}$$

$$= \frac{2}{1 + \frac{1}{2}}$$

1 mark

$$= \frac{4}{3}$$

1 mark



$$i \quad AD: \quad m_1 = \frac{-2-6}{2-0}$$

 $\frac{1}{2}$

$$= -4$$

$$AB: \quad m_2 = \frac{1}{4}$$

1 mark

$$\text{Eq'n AB: } (x_1, y_1) \quad m = \frac{1}{4}$$

$$y - 6 = \frac{1}{4}(x - 0) \quad 1 \text{ mark}$$

$$4y - 24 = x$$

$$\therefore 0 = x - 4y + 24$$

$$ii \quad B \text{ lies on } y=x: \quad 0 = x - 4y + 24$$

 $\frac{1}{2}$

$$= x - 4x + 24$$

1 mark

$$= -3x + 24$$

$$x = 8$$

1 mark

$$\text{Coordinates of B: } (8, 8)$$

Q12

A(0,6) B(8,8) D(2,-2)

(5)

iii
Area of trapezium = $\frac{1}{2}(a+b)$

$$AD = \sqrt{(2-0)^2 + (-2-6)^2}$$

$$= \sqrt{68}$$

$$\text{Area ABCD} = \frac{\sqrt{68}}{2} (\sqrt{68} + 2\sqrt{68})$$

$$= \frac{2\sqrt{17}}{2} (2\sqrt{17} + 4\sqrt{17})$$

$$= 34 + 68$$

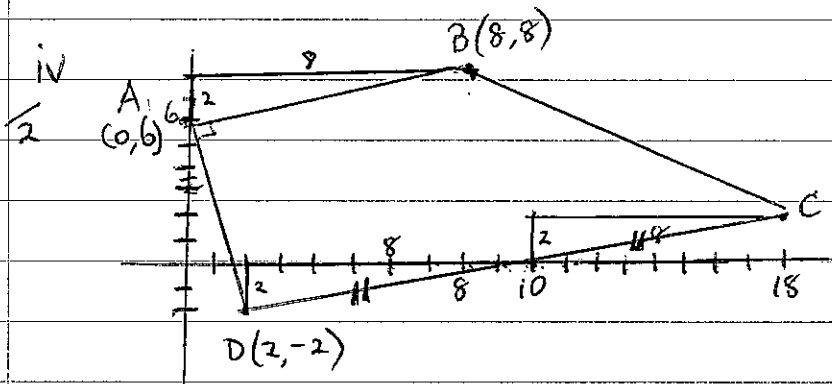
$$= 102 \text{ units}^2 \quad 1 \text{ mark}$$

$$AB = \sqrt{(8-0)^2 + (8-6)^2}$$

$$= \sqrt{68}$$

$$DC = 2 \times AB = 2\sqrt{68}$$

$$= 4\sqrt{17} \quad 1 \text{ mark}$$



Coordinates of C : (18, 2) 2 marks

c) i $y = x^2 + 2x - 5$ $y = -2x$

$\therefore x^2 + 2x - 5 = -2x$ 1 mark

$$x^2 + 4x - 5 = 0$$

$$(x+5)(x-1) = 0$$

$$x = -5 \text{ and } 1 \quad 1 \text{ mark}$$

x-coordinate of A = -5

x-coordinate of B = 1

ii $A_{-5}^1 = \int_{-5}^1 -2x - (x^2 + 2x - 5) dx$ 1 mark 1 mark

$$= \int_{-5}^1 -x^2 - 4x + 5 dx$$

$$= \left[-\frac{x^3}{3} - 2x^2 + 5x \right]_{-5}^1$$

$$= \left(-\frac{1}{3} - 2 + 5 \right) - \left(\frac{125}{3} - 50 - 25 \right)$$

$$= 36 \text{ units}^2 \quad 1 \text{ mark}$$

6

Question 13 (15 marks)

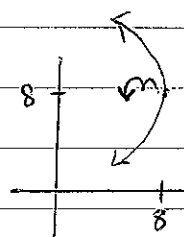
a) i $8x = 16y - y^2$
 $y^2 - 16y + 8^2 = -8x + 8^2$
 $(y-8)^2 = -8x + 64$
 $(y-8)^2 = -8(x-8)$
 $(y-h)^2 = -4a(x-h)$

1 mark

$\therefore$ vertex = (8, 8)

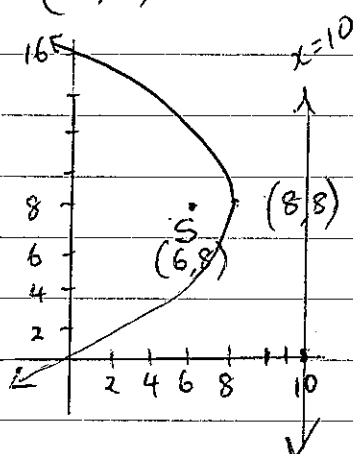
1 mark

ii focal length = 2
 focus = (6, 8)



1 mark

iii



$x=0 \rightarrow (y-8)^2 = 64$
 $y-8 = \pm 8$
 $y = 16, 0$

1 mark

b) i $V = 10000e^{-pt}$
 $V = V_0e^{-kt}$
 $\therefore V_0 = 10000$

$\therefore$ Value of the motorcycle is \$10000 when the man bought it.

1 mark

ii Value after 12 months is \$4000

$\rightarrow t = 1 \text{ year} ; V = 4000 : 4000 = 10000e^{-p}$

1 mark

$\frac{2}{5} = e^{-p}$

$\frac{5}{2} = e^p$

$\ln 2.5 = p$

$p = 0.916291$ (6 dp)

When $t = 1.5$ years: $V = 10000e^{-(0.916291 \times 1.5)}$
 $= 2529.8221 \dots$

1 mark

$\therefore$ Expected value after 18 months = \$2530 (nearest \$)

1 mark

Q13

(7)

$$b) \text{ iii } V = 10000 e^{-(\ln 2.5)t}$$

$$\frac{1}{2} \quad 1000 > 10000 e^{-(\ln 2.5)t}$$

$$\frac{1}{10} > e^{-(\ln 2.5)t}$$

$$\frac{1}{10} < e^{(\ln 2.5)t}$$

1 mark

$$\ln 10 < (\ln 2.5)t$$

$$\frac{\ln 10}{\ln 2.5} < t$$

$$\frac{\ln 10}{\ln 2.5}$$

$$i. \quad t > 2.51294...$$

$\therefore$ Age of motorcycle when value is less than \$1000

$$= 2 \text{ yrs } 7 \text{ months } (0.6 \text{ of } 12 \text{ mths} = 7.2 \text{ mths})$$

1 mark

$$c) \text{ i } v = \frac{12}{(t+1)^2} - 3$$

$$a = \frac{dv}{dt}$$

$$V = 12(t+1)^{-2} - 3$$

$$\frac{dv}{dt} = -24(t+1)^{-3} \cdot 1$$

$$\therefore a = \frac{-24}{(t+1)^3} \text{ m s}^{-2}$$

1 mark numerator

1 mark denominator

$$ii \quad v = 12(t+1)^{-2} - 3$$

$$\frac{1}{3} \quad x = \frac{12}{(t+1)^{-1}} - 3t + c$$

$$x = \frac{-12}{t+1} - 3t + c$$

$$\text{When } t=0, x=0: \quad 0 = -\frac{12}{1} - 0 + c$$

$$c = 12$$

$$\therefore x = \frac{-12}{t+1} - 3t + 12$$

(continued over page)

Q13

⑧

c) ii cont!

At rest when $v=0$: $0 = \frac{12}{(t+1)^2} - 3$

$$0 = 12 - 3(t+1)^2$$

$$-12 = -3(t+1)^2$$

$$4 = (t+1)^2$$

$$t+1 = \pm 2$$

$$t = 1 \text{ s or } -3 \text{ s}$$

1 mark

↑
Disregard -ve time
($t \geq 0$)

When $t=0$, $x=0$ m

When $t=1$, $x = \frac{-12}{(1)+1} - 3(1) + 12$

$$= 3 \text{ m}$$

$\therefore$ Distance travelled is 3 metres.

1 mark

Question 14 (15 marks)

a) i $x \mid 0 \ 1 \ 2$ $f(x) = \sqrt{16-x^2}$

$\frac{1}{2}$ $f(x) \mid 4 \ 2\sqrt{3}$

$$\int_a^b f(x) dx \doteq \frac{b-a}{2} (f(a) + f(b))$$

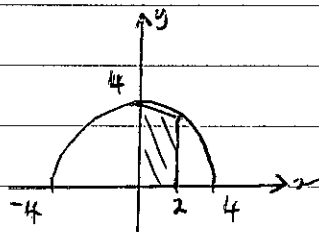
$$\int_0^2 \sqrt{16-x^2} dx \doteq \frac{2-0}{2} (\sqrt{16} + \sqrt{12}) \quad 1 \text{ mark}$$

$$\doteq 4 + 2\sqrt{3}$$

$$\doteq 7.46 \quad (2 \text{ d.p.}) \quad 1 \text{ mark}$$

ii

1



This approximation is less than the exact value as the trapezium formed by joining $f(0)$ to $f(2)$ lies below the curve. 1 mark

b) i $f(x) = x^3 + 3x^2 - 9x + 5$

$\frac{1}{3}$ $f'(x) = 3x^2 + 6x - 9$

$$f''(x) = 6x + 6$$

Stat pts when $f'(x) = 0$

$$\text{i.e. } 3x^2 + 6x - 9 = 0 \quad 1 \text{ mark}$$

$$x^2 + 2x - 3 = 0$$

$$(x+3)(x-1) = 0$$

$$x = -3 \text{ or } 1$$

$(-3, 32)$ $x \mid -4 \ -3 \ -2$ is a MAX or $f''(-3) = -12$

$$f'(x) \mid 15 \ 0 \ -9$$

$$\diagup \quad - \quad \diagdown$$

$$\therefore (-3, 32) \text{ is a Max}$$

1 mark

$(1, 0)$ $x \mid 0 \ 1 \ 2$ is a MIN or $f''(1) = 12$

$$f'(x) \mid -9 \ 0 \ 15$$

$$\diagdown \quad - \quad \diagup$$

$$\therefore (1, 0) \text{ is a Min}$$

1 mark

$\therefore$ Stat pts at $(-3, 32)$ MAX and $(1, 0)$ MIN

Q14

(10)

b) ii inflexion pt when $f''(x)=0$ $\frac{1}{2}$

$$6x+6=0$$

1 mark

$$x=-1$$

(-1, 16)

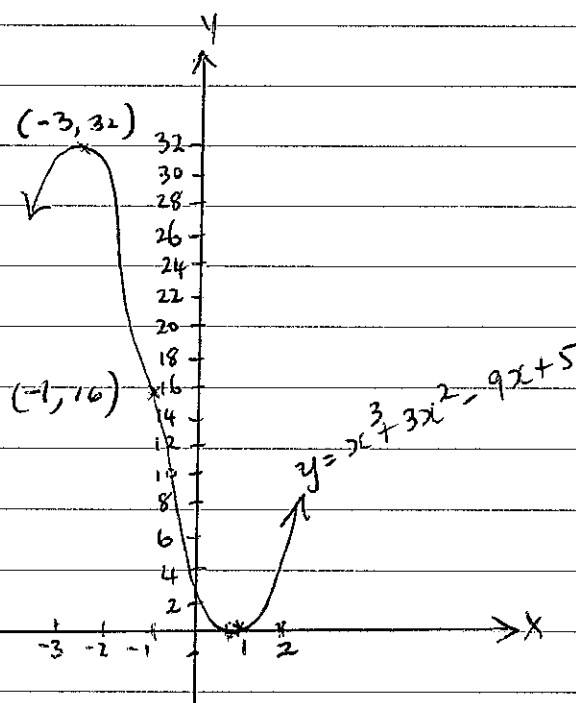
x	-2	-1	0
$f''(x)$	-6	0	6
	$\cap$	$\cup$	

→ change in concavity

∴ Pt of inflexion at (-1, 16)

1 mark

iii

 $\frac{1}{2}$ 

2 marks

c) i 6000, 6900, 7935, 9125.25, ...

 $\frac{1}{1}$

$$a=6000$$

$$r=1.15$$

$$T_n = ar^{n-1}$$

$$T_{10} = 6000 \times 1.15^9$$

$$= 21107.25775$$

$$\approx 21110 \text{ (nearest ten)}$$

1 mark

Sol'n: 21110 mobile phones sold in the 10th year. $\frac{1}{2}$

$$ii \quad S_n = \frac{a(r^n - 1)}{r - 1}, \quad r > 1$$

$$a = 6000$$

$$r = 1.15$$

$$S_{10} = \frac{6000(1.15^{10} - 1)}{0.15}$$

1 mark

$$= 121822.3094$$

1 mark.

Sol'n: Total sales in first 10 yrs = 121820 (nearest ten)

Q14

(11)

e) iii $\$1\,000\,000 \div \$10 = 100\,000$ sales $\frac{1}{2}$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$100\,000 = \frac{6000(1.15^n - 1)}{0.15}$$

1 mark

$$2.5 = 1.15^n - 1$$

$$3.5 = 1.15^n$$

$$\log 3.5 = n \log 1.15$$

$$n = \frac{\log 3.5}{\log 1.15}$$

$$= 8.963\,548\,64\dots$$

 $\therefore$ In the 9th year sales will reach 100 000 $\therefore$ profits will reach \$1 000 000.

1 mark

Question 15 (15 marks)

a) $\sin x = \cos x \quad 0 \leq x \leq 2\pi$

$\frac{1}{2} \quad \frac{\sin x}{\cos x} = 1$

$\tan x = 1$

$x = 45^\circ \text{ or } \frac{\pi}{4}$

$\frac{S}{T} \mid \frac{A}{C}$

$\tan x$ is positive, x is in 1st & 3rd quadrants

$\therefore x = \frac{\pi}{4}, \pi + \frac{\pi}{4}$

$\therefore x = \frac{\pi}{4}, \frac{5\pi}{4}$

1 mark for each solution

b) Prove $\frac{1}{1 + \tan^2 A} = (1 + \sin A)(1 - \sin A)$

$\frac{1}{2}$

LHS = $\frac{1}{1 + \tan^2 A}$

$[\tan^2 A + 1 = \sec^2 A]$

$= \frac{1}{1 + \sec^2 A - 1}$

$= \frac{1}{\sec^2 A}$

$= \cos^2 A$

$[\sin^2 A + \cos^2 A = 1]$

1 mark

$= 1 - \sin^2 A$

$= (1 + \sin A)(1 - \sin A)$

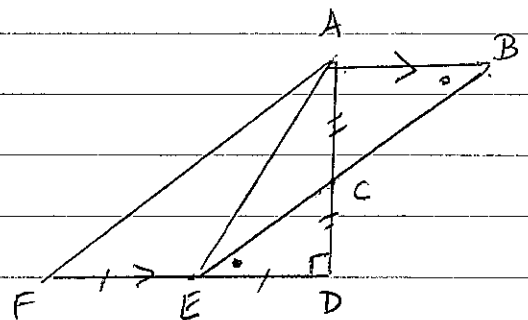
$= \text{RHS}$

1 mark

c) i $\angle CED = \angle ABC$

$\therefore$ Alternate angles equal;

$AB \parallel FD$



$\frac{1}{2}$ ii Prove $\triangle CDE \equiv \triangle CAB$

In Δ 's CDE and CAB: (A) $\angle CED = \angle ABC$ (Alternate angles equal; $AB \parallel FD$)

(A) $\angle ACB = \angle DCE$ (Vertically opposite angles)

(S) $AC = DC$ (given C is midpt of AB)

(1) - 2 correct reasons

(2) - All correct

$\therefore \triangle CDE \equiv \triangle CAB$ by AAS

Q15

c) iii Show $AF = 2BC$ 1/2 In Δ 's ADF and CDE:

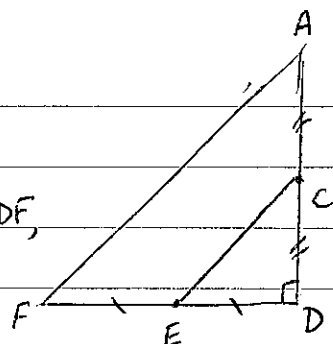
$$\frac{ED}{FD} = \frac{CD}{AD} = \frac{1}{2} \text{ (given E is midpt DF, C is midpt AD)}$$

 $\angle D$ is common

$\therefore \Delta ADF \parallel \Delta CDE$ (2 pairs of corresponding sides in same ratio and included angle is equal) 1 mark

$$\therefore AF = 2 \times CE$$

$\therefore AF = 2 \times BC$ (from ii - corresponding sides CE and BC in congruent triangles CDE and CAB) 2 marks



$$d) 2x^2 + 4x + 5 = 0$$

$$1 \quad i \quad \alpha + \beta = -\frac{b}{a} \quad a=2, b=4, c=5$$

$$= -\frac{4}{2}$$

$$= -2$$

1 mark

$$1/2 \quad ii \quad \text{Show } \frac{\alpha^2 + \beta^2}{\alpha\beta} = -\frac{2}{5}$$

$$LHS = \frac{\alpha^2 + \beta^2}{\alpha\beta}$$

$$= \frac{\alpha^2 + 2\alpha\beta + \beta^2 - 2\alpha\beta}{\alpha\beta}$$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta}$$

$$= \frac{(-2)^2 - 2\left(\frac{5}{2}\right)}{\frac{5}{2}}$$

$$= -1 \div \frac{5}{2}$$

$$= -\frac{2}{5}$$

$$= RHS$$

$$\alpha + \beta = -2$$

$$\alpha\beta = \frac{c}{a} = \frac{5}{2}$$

1 mark

1 mark

Q15

(14)

$$d) iii) x^2 + ax + b = 0$$

$$x^2 + (\alpha + \beta)x + (\alpha\beta) = 0$$

$$\text{Roots: } \frac{\alpha}{\beta} + 2, \frac{\beta}{\alpha} + 2$$

$$x^2 + mx + p = 0$$

$$\frac{\alpha^2 + \beta^2}{\alpha\beta} = -\frac{2}{5}$$

$$\frac{\alpha}{\beta} + 2 \quad \frac{\beta}{\alpha} + 2$$

$$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} + 4 = -m$$

$$\left(\frac{\alpha}{\beta} + 2\right)\left(\frac{\beta}{\alpha} + 2\right) = \frac{21}{5}$$

$$\left(\frac{\alpha}{\beta} + 2\right)\left(\frac{\beta}{\alpha} + 2\right) = p$$

$$\frac{2\alpha}{\beta} + \frac{2\beta}{\alpha} = -\frac{4}{5}$$

$$1 + \frac{2\alpha}{\beta} + \frac{2\beta}{\alpha} + 4 = p$$

$$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = -\frac{2}{5}$$

$$\frac{2\alpha^2 + 2\beta^2}{\alpha\beta} = p - 5$$

$$\therefore -\frac{2}{5} + 4 = -m$$

$$2 \times -\frac{2}{5} = p - 5$$

$$m = -\frac{18}{5}$$

$$p = 5 - \frac{4}{5}$$

$$= \frac{21}{5}$$

1 mark

1 mark.

Question 16 (15 marks)

a) i $D = \frac{t^2+1}{e^{2t}} \quad t \geq 0$

$$= \frac{u}{v} \quad \text{where } u = t^2+1, \quad v = e^{2t}$$

$$u' = 2t, \quad v' = 2e^{2t}$$

$$D' = \frac{vu' - uv'}{v^2}$$

$$= \frac{e^{2t} \cdot 2t - (t^2+1) \cdot 2e^{2t}}{(e^{2t})^2} \quad 1 \text{ mark}$$

$$= \frac{e^{2t} (2t - 2t^2 - 2)}{e^{4t}}$$

$$= \frac{-2(t^2 - t + 1)}{e^{2t}} \quad 1 \text{ mark}$$

ii $t = 10: D' = \frac{-2(10^2 - 10 + 1)}{e^{20}}$

$$= - \frac{182}{e^{20}}$$

$$\approx -3.75 \times 10^{-7}$$

$$< 0$$

$\therefore$ Depth of liquid decreasing at $t=10$
1 mark

b) i $y = x^2 \log_e x \quad x > 0$

$$= u \cdot v \quad \text{where } u = x^2, \quad v = \log_e x$$

$$u' = 2x, \quad v' = \frac{1}{x}$$

$$y' = uv' + vu'$$

$$= x^2 \cdot \frac{1}{x} + \log_e x \cdot 2x \quad 1 \text{ mark}$$

$$= x + 2x \log_e x \quad 1 \text{ mark}$$

$$= x(1 + 2 \log_e x)$$

Q16

b) ii $\int x \log_e x \, dx$

$\frac{1}{2} \int (2x + 2x \ln x) \, dx = x^2 \ln x$

$\int 2x \, dx + 2 \int x \ln x \, dx = x^2 \ln x$ 1 mark

$2 \int x \ln x \, dx = x^2 \ln x - \int x \, dx$

$\therefore \int x \ln x \, dx = \frac{x^2}{2} \ln x - \frac{x^2}{2} + C$ 1 mark

c) i $A = 2rl - \frac{\pi r^2}{2}$

$400 = 2rl - \frac{\pi r^2}{2}$ 1 mark

$800 = 4rl - \pi r^2$

$4rl = 800 + \pi r^2$

$l = \frac{800}{4r} + \frac{\pi r^2}{4r}$ 1 mark

$l = \frac{200}{r} + \frac{\pi}{4} r$

ii $P = \frac{2\pi r}{2} + 2l + 2r$

$= \pi r + 2r + 2 \left(\frac{200}{r} + \frac{\pi}{4} r \right)$ 1 mark

$= \pi r + 2r + \frac{400}{r} + \frac{\pi}{2} r$

$= r \left(\pi + 2 + \frac{\pi}{2} \right) + \frac{400}{r}$

$\therefore P = \left(\frac{3\pi}{2} + 2 \right) r + \frac{400}{r}$ 1 mark

iii $\frac{dP}{dr} = \frac{3\pi}{2} + 2 + (-2) \cdot 400 r^{-2}$

$= \frac{3\pi}{2} + 2 - \frac{800}{r^2}$

c) iii cont.

For stat pt, $\frac{dP}{dr} = 0$

$$\therefore \frac{3\pi}{2} + 2 = \frac{400}{r^2} \quad (1)$$

$$\frac{3\pi + 4}{2} = \frac{400}{r^2}$$

$$r^2 = \frac{800}{3\pi + 4}$$

$$r = \pm \sqrt{\frac{800}{3\pi + 4}}$$

$r > 0$ since r is a length

$$\therefore r = \sqrt{\frac{800}{3\pi + 4}} \doteq 7.72 \text{ m (2 dp)} \quad (1)$$

$$\begin{aligned} \frac{d^2P}{dr^2} &= -2(-400r^{-3}) \\ &= \frac{800}{r^3} \end{aligned}$$

$$\begin{aligned} \text{When } r = 7.72, \quad \frac{d^2P}{dr^2} &= \frac{800}{7.72^3} \\ &> 0 \end{aligned} \quad (1)$$

$\therefore P$ is a maximum when $r \doteq 7.72 \text{ m}$